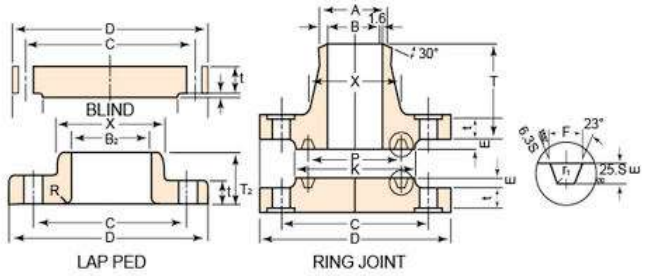
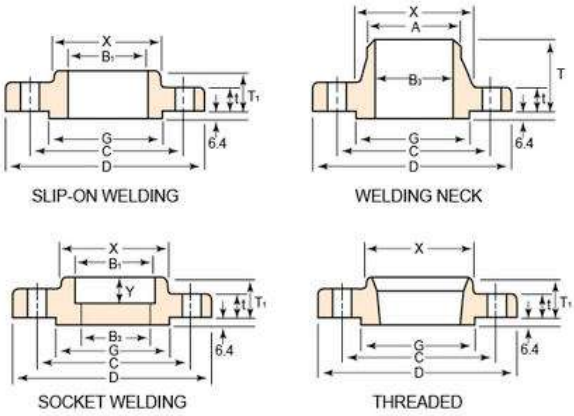
# ANSI B16.5




## CLASS 400

Unit : mm

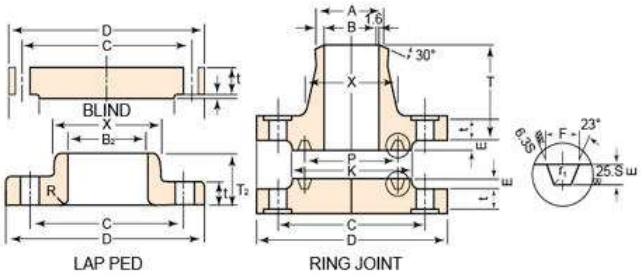
| Nominal Pipe Size | | Outside Dia. of Flanges | Dia. of Bore | | | | | | | | | | Dia. or Hub at Bevel A | | Dia. of Hub at Base | Dia. of Raised Face | thick of Flanges | Radius of Fillet |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Slip-on Socket $B_1$ | | Lap-Joint $B_2$ | | Welding-Neck, Socket $B_3$ | | | | | | | | | | | |
| | | | | | | | JPI | | | ANSI | | | | | | | | |
| A | B | D | JPI | ANSI | JPI | ANSI | SCH40 | SCH80 | SCH160 | SCH40 | SCH80 | SCH160 | JPI | ANSI | X | G | t | R |
| 15 | ½ | 95.3 | 22.2 | 22.4 | - | 22.9 | 16.1 | 14.3 | 12.3 | 15.8 | 13.9 | 11.8 | 21.7 | 21.3 | 38.1 | 34.9 | 14.3 | 3 |
| 20 | ¾ | 117.3 | 27.8 | 27.7 | - | 28.2 | 21.4 | 19.4 | 16.2 | 20.9 | 18.8 | 15.6 | 27.2 | 26.7 | 47.8 | 42.9 | 15.9 | 3 |
| 25 | 1 | 124.0 | 34.5 | 34.6 | - | 35.1 | 27.2 | 25.0 | 21.2 | 26.6 | 24.3 | 20.7 | 34.0 | 33.5 | 53.8 | 50.8 | 17.5 | 3 |
| (32) | (1¼) | 133.4 | 43.2 | 43.2 | - | 43.7 | 35.5 | 32.9 | 29.9 | 35.1 | 32.5 | 29.5 | 42.7 | 42.2 | 63.5 | 63.5 | 20.7 | 5 |
| 40 | 1½ | 155.4 | 49.1 | 49.6 | - | 50.0 | 41.2 | 38.4 | 34.4 | 40.9 | 38.1 | 34.0 | 48.6 | 48.3 | 69.9 | 73.0 | 22.3 | 6 |
| 50 | 2 | 165.1 | 61.1 | 62.0 | - | 62.5 | 52.7 | 49.5 | 43.1 | 52.5 | 49.3 | 42.9 | 60.5 | 60.5 | 84.1 | 92.1 | 25.4 | 8 |
| 65 | 2½ | 190.5 | 77.1 | 74.7 | - | 75.5 | 65.9 | 62.3 | 57.3 | 62.7 | 59.0 | 54.0 | 76.3 | 73.2 | 100.1 | 104.8 | 28.6 | 8 |
| 80 | 3 | 209.6 | 90.0 | 90.7 | - | 91.4 | 78.1 | 73.9 | 66.9 | 77.9 | 73.7 | 66.7 | 89.1 | 88.9 | 117.3 | 127.0 | 31.8 | 10 |
| (90) | (3½) | 228.6 | 102.6 | 103.4 | - | 104.2 | 90.2 | 85.4 | 76.2 | 90.1 | 85.5 | - | 101.6 | 101.6 | 133.4 | 139.7 | 35.0 | 10 |
| 100 | 4 | 254.0 | 115.4 | 116.1 | - | 116.9 | 102.3 | 97.1 | 87.3 | 102.3 | 97.2 | 87.3 | 114.3 | 114.3 | 146.1 | 157.2 | 35.0 | 11 |
| (125) | (5) | 279.4 | 141.2 | 143.8 | - | 144.6 | 126.6 | 120.8 | 108.0 | 128.2 | 122.3 | 109.6 | 139.8 | 141.2 | 177.8 | 185.7 | 38.1 | 11 |
| 150 | 6 | 317.5 | 166.6 | 170.7 | - | 171.5 | 151.0 | 143.2 | 128.8 | 154.1 | 146.3 | 131.8 | 165.2 | 168.4 | 206.2 | 215.9 | 41.1 | 13 |
| 200 | 8 | 381.0 | 218.0 | 221.5 | - | 222.3 | 199.9 | 190.9 | 170.3 | 202.7 | 193.7 | 173.1 | 216.3 | 219.2 | 260.3 | 269.9 | 47.7 | 13 |
| 250 | 10 | 444.5 | 269.5 | 276.4 | - | 277.4 | 248.8 | 237.2 | 210.2 | 254.5 | 242.9 | 215.9 | 267.4 | 273.1 | 320.5 | 323.8 | 54.0 | 13 |
| 300 | 12 | 520.7 | 321.0 | 327.2 | - | 328.2 | 297.9 | 283.7 | 251.9 | 303.2 | 288.9 | 257.2 | 318.5 | 323.9 | 374.7 | 381.0 | 57.2 | 13 |
| 350 | 14 | 584.2 | 358.1 | 359.2 | - | 360.2 | 333.4 | 317.6 | 284.2 | 333.3 | 317.5 | 284.2 | 355.6 | 355.6 | 425.5 | 412.8 | 60.4 | 13 |
| 400 | 16 | 647.7 | 409.0 | 410.5 | - | 411.3 | 381.0 | 363.3 | 325.4 | 381.0 | 363.6 | 325.5 | 406.4 | 406.4 | 482.6 | 469.9 | 63.5 | 13 |
| 450 | 18 | 711.2 | 460.0 | 461.8 | - | 462.3 | 428.6 | 409.6 | 366.8 | 428.7 | 409.6 | 366.7 | 457.2 | 457.2 | 533.4 | 533.4 | 66.7 | 13 |
| 500 | 20 | 774.7 | 511.0 | 513.1 | - | 514.4 | 477.8 | 455.6 | 408.0 | 477.8 | 455.6 | 408.0 | 508.0 | 508.0 | 578.4 | 584.2 | 69.9 | 13 |
| 600 | 24 | 914.4 | 613.0 | 616.0 | - | 616.0 | 574.6 | 547.8 | 490.6 | 574.6 | 547.7 | 490.6 | 609.6 | 609.6 | 701.5 | 692.2 | 76.2 | 13 |

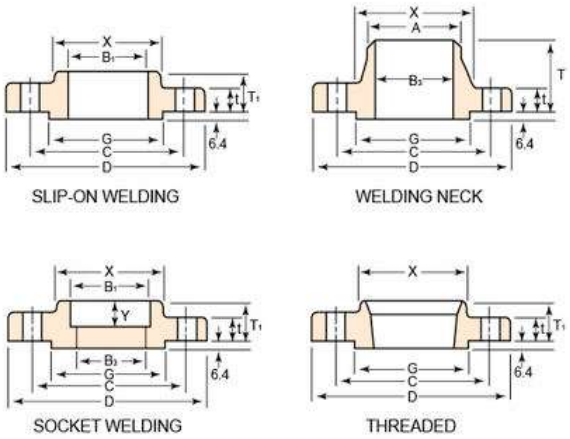
## CLASS 400

Unit : mm

| Nominal Pipe Size | | Length Thru Hub | | | Depth or Socket | Drilling | | | Dia. of Raised Face for Ring Joint | Pitch Dia. of Ring & Groove | Ring No | RTJ Type | | | Approx. Weight (kg) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Slip-on Socket Thread | Lap Joint | Welding Neck | | Dia. of Bolt Circle | Number or Bolt | Dia. of Bolt Hole | | | | Depth of Groove | Width of Groove | Radius of Groove | WN | SO | SW | BL |
| A | B | $T_1$ | $T_2$ | T | Y | C | N | | K (min) | P | | E | F | $r_1$ | | | | |
| 15 | ½ | 22.4 | 22 | 52.3 | 9.5 | 66.5 | 4 | 16 | 51.0 | 34.14 | R11 | 5.56 | 7.14 | 0.8 | 0.86 | 0.74 | 0.77 | 0.76 |
| 20 | ¾ | 25.4 | 25 | 57.2 | 11.1 | 82.6 | 4 | 19 | 63.5 | 42.88 | R13 | 6.35 | 8.74 | 0.8 | 1.40 | 1.24 | 1.28 | 1.27 |
| 25 | 1 | 26.9 | 27 | 62.0 | 12.7 | 88.9 | 4 | 19 | 70.0 | 50.80 | R16 | 6.35 | 8.74 | 0.8 | 1.76 | 1.51 | 1.57 | 1.59 |
| (32) | (1¼) | 28.4 | 29 | 66.5 | 14.3 | 98.6 | 4 | 19 | 79.5 | 60.32 | R18 | 6.35 | 8.74 | 0.8 | 2.41 | 2.03 | 2.11 | 2.24 |
| 40 | 1½ | 31.8 | 32 | 69.9 | 15.9 | 114.3 | 4 | 22 | 90.5 | 68.28 | R20 | 6.35 | 8.74 | 0.8 | 3.44 | 3.01 | 3.11 | 3.30 |
| 50 | 2 | 36.6 | 37 | 73.2 | 17.5 | 127.0 | 8 | 19 | 108 | 82.55 | R23 | 7.92 | 11.91 | 0.8 | 4.18 | 3.62 | 3.77 | 4.11 |
| 65 | 2½ | 41.1 | 41 | 79.2 | 19.0 | 149.4 | 8 | 22 | 127 | 101.60 | R26 | 7.92 | 11.91 | 0.8 | 6.11 | 5.20 | 5.48 | 6.20 |
| 80 | 3 | 46.0 | 46 | 82.6 | 20.6 | 168.1 | 8 | 22 | 146 | 123.82 | R31 | 7.92 | 11.91 | 0.8 | 8.27 | 7.08 | 7.47 | 8.51 |
| (90) | (3½) | 49.3 | 49 | 85.9 | 23.2 | 184.2 | 8 | 25 | 159 | 131.78 | R34 | 7.92 | 11.91 | 0.8 | 10.3 | 8.86 | - | 10.9 |
| 100 | 4 | 50.8 | 51 | 88.9 | 23.8 | 200.2 | 8 | 25 | 175 | 149.22 | R37 | 7.92 | 11.91 | 0.8 | 12.9 | 10.89 | - | 13.7 |
| (125) | (5) | 53.8 | 54 | 101.6 | 23.8 | 235.0 | 8 | 25 | 210 | 180.98 | R41 | 7.92 | 11.91 | 0.8 | 17.1 | 14.07 | - | 18.6 |
| 150 | 6 | 57.2 | 57 | 103.1 | 27.0 | 269.7 | 12 | 25 | 241 | 211.12 | R45 | 7.92 | 11.91 | 0.8 | 22.23 | 18.9 | - | 25.7 |
| 200 | 8 | 68.3 | 68 | 117.3 | 31.8 | 330.2 | 12 | 29 | 302 | 269.88 | R49 | 7.92 | 11.91 | 0.8 | 35.38 | 29.4 | - | 42.9 |
| 250 | 10 | 73.2 | 101.6 | 124.0 | 33.3 | 387.4 | 16 | 32 | 356 | 323.85 | R53 | 7.92 | 11.91 | 0.8 | 49.89 | 41.1 | - | 64.6 |
| 300 | 12 | 79.2 | 108.0 | 136.7 | 39.7 | 450.9 | 16 | 35 | 413 | 381.00 | R57 | 7.92 | 11.91 | 0.8 | 72.57 | 59.2 | - | 95.7 |
| 350 | 14 | 84.1 | 117.3 | 149.4 | 44.3 | 514.4 | 20 | 35 | 457 | 419.10 | R61 | 7.92 | 11.91 | 0.8 | 99.0 | 80.0 | - | 125 |
| 400 | 16 | 93.7 | 127 | 152.4 | 44.5 | 571.5 | 20 | 39 | 508 | 469.90 | R65 | 7.92 | 11.91 | 0.8 | 125 | 103 | - | 162 |
| 450 | 18 | 98 | 136.7 | 165.1 | 49.2 | 628.7 | 24 | 39 | 575 | 533.40 | R69 | 7.92 | 11.91 | 0.8 | 153 | 123 | - | 205 |
| 500 | 20 | 101.6 | 146.1 | 168.1 | 54.0 | 685.8 | 24 | 42 | 635 | 584.20 | R73 | 9.52 | 13.49 | 1.5 | 184 | 148 | - | 255 |
| 600 | 24 | 114.3 | 158.8 | 174.8 | 63.5 | 812.8 | 24 | 48 | 749 | 692.15 | R77 | 11.13 | 16.66 | 1.5 | 270 | 223 | - | 388 |

# ANSI B16.5



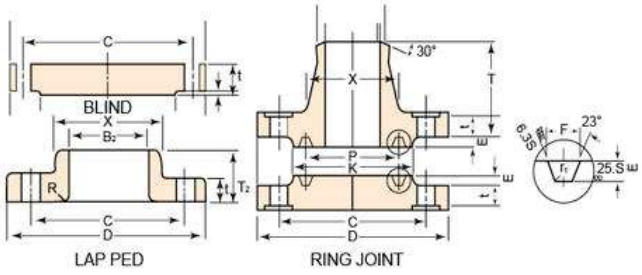



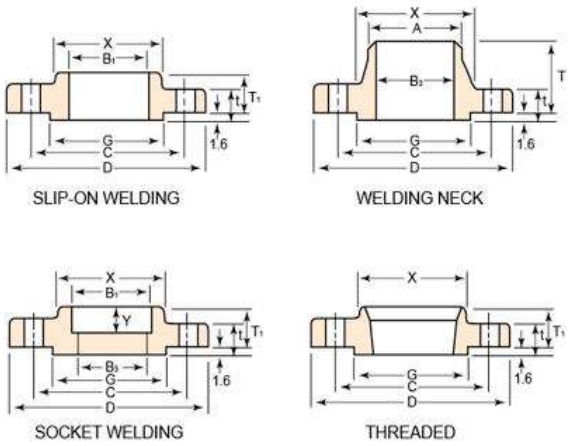

## CLASS 900

Unit : mm

| Nominal Pipe Size | | Length Thru Hub | | | Depth or Socket | Drilling | | | Dia. of Raised Face for Ring Joint | Pitch Dia. of Ring & Groove | Ring No | RTJ Type | | | Approx. Weight (kg) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Slip-on Socket Thread | Lap Joint | Welding Neck | | Dia. of Bolt Circle | Number or Botl | Dia. of Bolt Hole | | | | Depth of Groove | Width of Groove | Radius of Groove | WN | SO | SW | BL |
| A | B | $T_1$ | $T_2$ | T | Y | C | N | | K (min) | P | | E | F | $r_1$ | | | | |
| 15 | ½ | 31.8 | 31.8 | 60.5 | 9.5 | 82.6 | 4 | 23 | 60.5 | 39.67 | R12 | 6.35 | 8.74 | 0.8 | 1.98 | 1.76 | 1.81 | 1.88 |
| 20 | ¾ | 35.1 | 35.1 | 69.9 | 11.1 | 88.9 | 4 | 23 | 66.5 | 44.45 | R14 | 6.35 | 8.74 | 0.8 | 2.64 | 2.32 | 2.39 | 2.50 |
| 25 | 1 | 41.1 | 41.1 | 73.2 | 12.7 | 101.6 | 4 | 26 | 71.5 | 50.80 | R16 | 6.35 | 8.74 | 0.8 | 3.84 | 3.44 | 3.56 | 3.68 |
| (32) | (1¼) | 41.1 | 41.1 | 73.2 | 14.3 | 111.3 | 4 | 26 | 81.0 | 60.32 | R18 | 6.35 | 8.74 | 0.8 | 4.47 | 3.95 | 4.11 | 4.19 |
| 40 | 1½ | 44.5 | 44.5 | 82.6 | 15.9 | 124.0 | 4 | 29 | 92.0 | 68.28 | R20 | 6.35 | 8.74 | 0.8 | 5.90 | 5.41 | 5.61 | 5.92 |
| 50 | 2 | 57.2 | 57.2 | 101.6 | 17.5 | 165.1 | 8 | 26 | 124 | 95.25 | R24 | 7.92 | 11.91 | 0.8 | 11.3 | 9.92 | 10.3 | 10.5 |
| 65 | 2½ | 63.5 | 63.5 | 104.6 | 19 | 190.5 | 8 | 29 | 137 | 107.95 | R27 | 7.92 | 11.91 | 0.8 | 15.3 | 13.5 | 14.2 | 14.2 |
| 80 | 3 | 53.8 | 53.8 | 101.6 | 20.6 | 190.5 | 8 | 26 | 156 | 123.82 | R31 | 7.92 | 11.91 | 0.8 | 14.2 | 11.6 | - | 13.17 |
| (90) | (3½) | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 100 | 4 | 69.9 | 69.9 | 114.3 | 23.8 | 235.0 | 8 | 32 | 181 | 157.22 | R37 | 7.92 | 11.91 | 0.8 | 23.1 | 19.8 | - | 22.7 |
| (125) | (5) | 79.3 | 79.2 | 127.0 | 23.8 | 279.4 | 8 | 35 | 216 | 180.98 | R41 | 7.92 | 11.91 | 0.8 | 37.4 | 32.4 | - | 37.5 |
| 150 | 6 | 85.9 | 85.9 | 139.7 | 27.0 | 317.5 | 12 | 32 | 241 | 211.12 | R45 | 7.92 | 11.91 | 0.8 | 49.3 | 42.1 | - | 48.7 |
| 200 | 8 | 101.6 | 114.3 | 162.1 | 31.8 | 393.7 | 12 | 39 | 308 | 269.88 | R49 | 7.92 | 11.91 | 0.8 | 79.45 | 71.6 | - | 84.0 |
| 250 | 10 | 108.0 | 127.0 | 184.2 | 33.3 | 469.9 | 16 | 39 | 362 | 323.85 | R53 | 7.92 | 11.91 | 0.8 | 118.04 | 102 | - | 124 |
| 300 | 12 | 117.3 | 142.7 | 200.2 | 39.7 | 533.4 | 20 | 39 | 419 | 381.00 | R57 | 7.92 | 11.91 | 0.8 | 157.0 | 136 | - | 173 |
| 350 | 14 | 130.0 | 155.4 | 212.9 | 44.3 | 558.8 | 20 | 42 | 467 | 419.10 | R62 | 11.13 | 16.66 | 1.5 | 181.60 | 153 | - | 206 |
| 400 | 16 | 133.4 | 165.1 | 215.9 | 44.5 | 616.0 | 20 | 45 | 524 | 469.90 | R66 | 11.13 | 16.66 | 1.5 | 224.73 | 186 | - | 259 |
| 450 | 18 | 152.4 | 190.5 | 228.6 | 49.2 | 685.8 | 20 | 51 | 594 | 533.40 | R70 | 12.70 | 19.84 | 1.5 | 308.72 | 258 | - | 367 |
| 500 | 20 | 158.8 | 209.6 | 247.7 | 54.0 | 749.3 | 20 | 54 | 648 | 584.20 | R74 | 12.70 | 19.84 | 1.5 | 376.82 | 317 | - | 463 |
| 600 | 24 | 203.2 | 266.7 | 292.1 | 63.5 | 901.7 | 20 | 67 | 772 | 692.15 | R78 | 15.58 | 26.97 | 2.4 | 685 | 608 | - | 876 |

# ANSI B16.5




## CLASS 300

Unit : mm

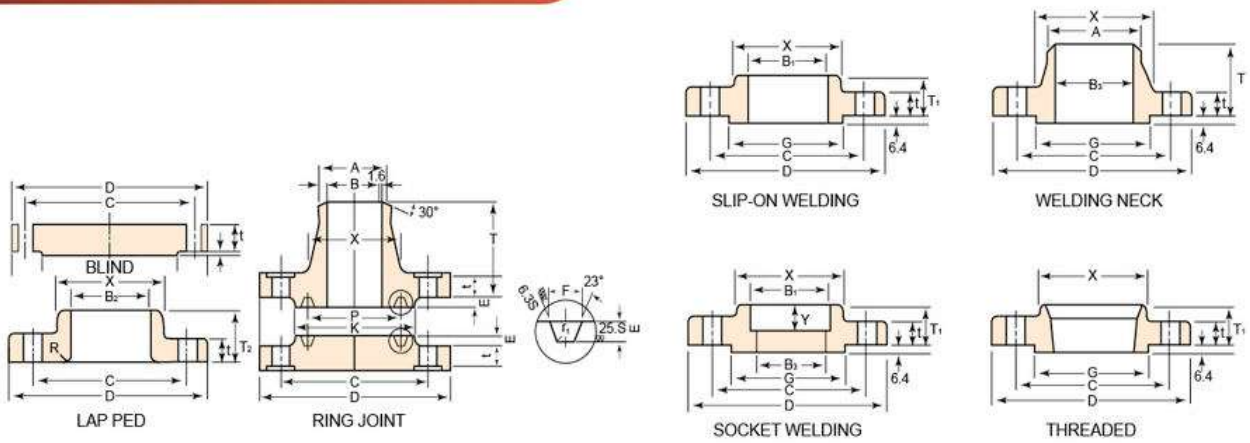
| Nominal Pipe Size | | Outside Dia. of Flanges | Dia. of Bore | | | | | | | | | | | Dia. or Hub at Bevel A | | Dia. of Hub at Base | Dia. of Raised Face | thick of Flanges | Radius of Fillet |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Slip-on Socket $B_1$ | | Lap-Joint $B_2$ | | Welding-Neck, Socket $B_3$ | | | | | | | | | | | | |
| | | | | | | | JPI | | | ANSI | | | | | | | | | |
| A | B | D | JPI | ANSI | JPI | ANSI | SCH40 | SCH80 | SCH160 | SCH40 | SCH80 | SCH160 | JPI | ANSI | X | G | t | R |
| 15 | ½ | 95.3 | 22.2 | 22.4 | 23.4 | 22.9 | 16.1 | 14.3 | 12.3 | 15.8 | 13.9 | 11.8 | 21.7 | 21.3 | 38.1 | 35.1 | 14.3 | 3 |
| 20 | ¾ | 117.3 | 27.7 | 27.7 | 28.9 | 28.2 | 21.4 | 19.4 | 16.2 | 20.9 | 18.8 | 15.6 | 27.2 | 26.7 | 47.6 | 42.9 | 15.9 | 3 |
| 25 | 1 | 124.0 | 34.5 | 34.6 | 35.6 | 35.1 | 27.2 | 25.0 | 21.2 | 26.6 | 24.3 | 20.7 | 34.0 | 33.5 | 54.0 | 50.8 | 17.5 | 3 |
| (32) | (1¼) | 133.4 | 43.2 | 43.2 | 44.3 | 43.7 | 35.5 | 32.9 | 29.9 | 35.1 | 32.5 | 29.5 | 42.7 | 42.2 | 63.5 | 63.5 | 19.1 | 5 |
| 40 | 1½ | 155.4 | 49.1 | 49.6 | 50.5 | 50.0 | 41.2 | 38.4 | 34.4 | 40.9 | 38.1 | 34.0 | 48.6 | 48.3 | 69.9 | 73.2 | 20.6 | 6 |
| 50 | 2 | 165.1 | 61.1 | 62.0 | 62.7 | 62.5 | 52.7 | 49.5 | 43.1 | 52.5 | 49.3 | 42.9 | 60.5 | 60.5 | 84.1 | 91.9 | 22.4 | 8 |
| 65 | 2½ | 190.5 | 77.1 | 74.7 | 78.7 | 75.5 | 65.9 | 62.3 | 57.3 | 62.7 | 59.0 | 54.0 | 76.3 | 73.2 | 100.0 | 104.6 | 25.4 | 8 |
| 80 | 3 | 209.6 | 90.0 | 90.7 | 91.6 | 91.4 | 78.1 | 73.9 | 66.9 | 77.9 | 73.7 | 66.7 | 89.1 | 88.9 | 117.5 | 127.0 | 28.6 | 10 |
| (90) | (3½) | 228.6 | 102.6 | 103.4 | 104.1 | 104.2 | 90.2 | 85.4 | 76.2 | 90.1 | 85.5 | - | 101.6 | 101.6 | 133.4 | 139.7 | 30.2 | 10 |
| 100 | 4 | 254.0 | 115.4 | 116.1 | 116.9 | 116.9 | 102.3 | 97.1 | 87.3 | 102.3 | 97.2 | 87.3 | 114.3 | 114.3 | 146.1 | 157.2 | 31.8 | 11 |
| (125) | (5) | 279.4 | 141.2 | 143.8 | 143.0 | 144.6 | 126.6 | 120.8 | 108.0 | 128.2 | 122.3 | 109.6 | 139.8 | 141.2 | 177.8 | 185.7 | 35.1 | 11 |
| 150 | 6 | 317.5 | 166.6 | 170.7 | 168.4 | 171.5 | 151.0 | 143.2 | 128.8 | 154.1 | 146.3 | 131.8 | 165.2 | 168.4 | 206.4 | 215.9 | 36.6 | 13 |
| 200 | 8 | 381.0 | 218.0 | 221.5 | 219.5 | 222.3 | 199.9 | 190.9 | 170.3 | 202.7 | 193.7 | 173.1 | 216.3 | 219.2 | 260.4 | 269.7 | 41.3 | 13 |
| 250 | 10 | 444.5 | 269.5 | 276.4 | 271.7 | 277.4 | 248.8 | 237.2 | 210.2 | 254.5 | 242.9 | 215.9 | 267.4 | 273.1 | 320.7 | 323.9 | 47.8 | 13 |
| 300 | 12 | 520.7 | 321.0 | 327.2 | 322.8 | 328.2 | 297.9 | 283.7 | 251.9 | 303.2 | 288.9 | 257.2 | 318.5 | 323.9 | 374.7 | 381.0 | 50.8 | 13 |
| 350 | 14 | 584.2 | 358.1 | 359.2 | - | 360.2 | 333.4 | 317.6 | 284.2 | 333.3 | 317.5 | 284.2 | 355.6 | 355.6 | 425.5 | 412.8 | 54.0 | 13 |
| 400 | 16 | 647.7 | 409.0 | 410.5 | - | 411.3 | 381.0 | 363.6 | 325.4 | 381.0 | 363.6 | 325.5 | 406.4 | 406.4 | 482.6 | 469.9 | 57.2 | 13 |
| 450 | 18 | 711.2 | 460.0 | 461.8 | - | 462.3 | 428.6 | 409.6 | 366.8 | 428.7 | 409.6 | 366.7 | 457.2 | 457.2 | 533.4 | 533.4 | 60.4 | 13 |
| 500 | 20 | 774.7 | 511.0 | 513.1 | - | 514.4 | 477.8 | 455.6 | 408.0 | 477.8 | 455.6 | 408.0 | 508.0 | 508.0 | 587.4 | 584.2 | 63.5 | 13 |
| 600 | 24 | 914.4 | 613.0 | 616.9 | - | 616.0 | 574.6 | 574.8 | 490.6 | 574.7 | 574.7 | 490.6 | 609.6 | 609.6 | 701.7 | 692.2 | 69.9 | 13 |

## CLASS 300

Unit : mm

| Nominal Pipe Size | | Length Thru Hub | | | Depth or Socket | Drilling | | | Dia. of Raised Face for Ring Joint | Pitch Dia. of Ring & Groove | Ring No | RTJ Type | | | Approx. Weight (kg) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Slip-on Socket Thread | Lap Joint | Welding Neck | | Dia. of Bolt Circle | Number or Bolt | Dia. of Bolt Hole | | | | Depth of Groove | Width of Groove | Radius of Groove | WN | SO | SW | BL |
| A | B | $T_1$ | $T_2$ | T | Y | C | N | | K (min) | P | | E | F | $r_1$ | | | | |
| 15 | ½ | 22.2 | 22.2 | 52.4 | 9.7 | 66.5 | 4 | 16 | 51.0 | 34.14 | R11 | 5.56 | 7.14 | 0.8 | 0.80 | 0.65 | 0.62 | 0.65 |
| 20 | ¾ | 25.4 | 25.4 | 57.2 | 11.2 | 82.6 | 4 | 19 | 63.5 | 42.88 | R13 | 6.35 | 8.74 | 0.8 | 1.25 | 1.10 | 1.12 | 1.09 |
| 25 | 1 | 27.0 | 27.0 | 61.9 | 12.7 | 88.9 | 4 | 19 | 70.0 | 50.80 | R16 | 6.35 | 8.74 | 0.8 | 1.58 | 1.35 | 1.39 | 1.38 |
| (32) | (1¼) | 27.0 | 27.0 | 65.1 | 14.2 | 98.6 | 4 | 19 | 79.5 | 60.32 | R18 | 6.35 | 8.74 | 0.8 | 2.05 | 1.69 | 1.74 | 1.82 |
| 40 | 1½ | 30.2 | 30.2 | 68.3 | 15.8 | 114.3 | 4 | 22 | 90.5 | 68.28 | R20 | 6.35 | 8.74 | 0.8 | 2.93 | 2.54 | 2.60 | 2.70 |
| 50 | 2 | 33.3 | 33.3 | 69.8 | 17.5 | 127.0 | 8 | 19 | 108 | 82.55 | R23 | 7.92 | 11.91 | 0.8 | 3.40 | 2.8 | 2.94 | 3.09 |
| 65 | 2½ | 38.1 | 38.1 | 76.2 | 19.1 | 149.4 | 8 | 22 | 127 | 101.60 | R26 | 7.92 | 11.91 | 0.8 | 5.10 | 4.24 | 4.43 | 4.75 |
| 80 | 3 | 42.9 | 42.9 | 79.4 | 20.6 | 168.1 | 8 | 22 | 146 | 123.22 | R31 | 7.92 | 11.91 | 0.8 | 7.01 | 5.8 | 6.21 | 6.79 |
| (90) | (3½) | 44.5 | 44.5 | 81.0 | 22.4 | 184.2 | 8 | 22 | 159 | 131.78 | R34 | 7.92 | 11.91 | 0.8 | 8.71 | 7.7 | - | 8.79 |
| 100 | 4 | 47.6 | 47.6 | 85.7 | 23.9 | 200.2 | 8 | 22 | 175 | 149.02 | R37 | 7.92 | 11.91 | 0.8 | 11.3 | 9.71 | - | 11.6 |
| (125) | (5) | 50.8 | 50.8 | 98.4 | 23.9 | 235.0 | 8 | 22 | 210 | 180.98 | R41 | 7.92 | 11.91 | 0.8 | 15.1 | 12.58 | - | 15.5 |
| 150 | 6 | 52.4 | 52.4 | 98.4 | 26.9 | 269.7 | 12 | 22 | 241 | 211.12 | R45 | 7.92 | 11.91 | 0.8 | 19.6 | 16.2 | - | 21.2 |
| 200 | 8 | 61.9 | 61.9 | 111.1 | 31.8 | 330.2 | 12 | 26 | 302 | 269.88 | R49 | 7.92 | 11.91 | 0.8 | 30.3 | 24.8 | - | 34.5 |
| 250 | 10 | 66.7 | 95.3 | 117.5 | 33.3 | 387.3 | 16 | 29 | 356 | 323.85 | R53 | 7.92 | 11.91 | 0.8 | 43.74 | 34.16 | - | 53.9 |
| 300 | 12 | 73.0 | 101.6 | 130.2 | 39.6 | 450.9 | 16 | 32 | 413 | 381.00 | R57 | 7.92 | 11.91 | 0.8 | 64.1 | 51 | - | 78.9 |
| 350 | 14 | 76.2 | 111.3 | 142.9 | 41.4 | 514.4 | 20 | 32 | 457 | 419.10 | R61 | 7.92 | 11.91 | 0.8 | 88.3 | 70.1 | - | 106 |
| 400 | 16 | 82.6 | 120.7 | 146.0 | 44.5 | 571.5 | 20 | 35 | 508 | 469.90 | R65 | 7.92 | 11.91 | 0.8 | 113 | 90.4 | - | 139 |
| 450 | 18 | 88.9 | 130.0 | 158.8 | 49.3 | 628.7 | 24 | 35 | 575 | 533.40 | R69 | 7.92 | 11.91 | 0.8 | 138 | 109 | - | 175 |
| 500 | 20 | 95.2 | 139.7 | 161.9 | 54.1 | 685.8 | 24 | 35 | 635 | 584.20 | R73 | 9.52 | 13.49 | 1.5 | 161.4 | 136 | - | 222 |
| 600 | 24 | 106.4 | 152.4 | 168.3 | 63.5 | 812.8 | 24 | 42 | 749 | 692.15 | R77 | 11.13 | 16.66 | 1.5 | 235.4 | 204 | - | 340 |

# ANSI B16.5



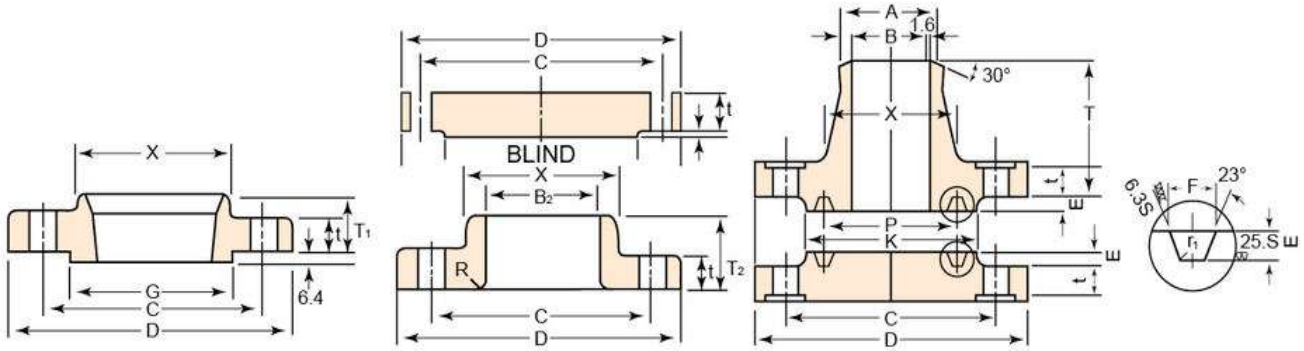

## CLASS 900

### ANSI & JPI FORGED FLANGES (JPI 7S-15-84 & ANSI B16.5)

Unit : mm

| Nominal Pipe Size | | Outside Dia. of Flanges | Dia. of Bore | | | | | | | | | | Dia. or Hub at Bevel A | | Dia. of Hub at Base | Dia. of Raised Face | thick of Flanges | Radius of Fillet |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Slip-on Socket $B_3$ | | Lap-Joint $B_2$ | | Welding-Neck, Socket $B_3$ | | | | | | | | | | | |
| | | | | | | | JPI | | | ANSI | | | | | | | | |
| A | B | D | JPI | ANSI | JPI | ANSI | SCH40 | SCH80 | SCH160 | SCH40 | SCH80 | SCH160 | JPI | ANSI | X | G | t | R |
| 15 | ½ | 120.7 | 22.2 | 22.4 | - | 22.9 | 16.1 | 14.3 | 12.3 | 15.8 | 13.9 | 11.8 | 21.7 | 21.3 | 38.1 | 34.9 | 22.4 | 3 |
| 20 | ¾ | 130 | 27.7 | 27.7 | - | 28.2 | 21.4 | 19.4 | 16.2 | 20.9 | 18.8 | 15.6 | 27.2 | 26.7 | 44.5 | 42.9 | 25.4 | 3 |
| 25 | 1 | 149.4 | 34.5 | 34.5 | - | 35.1 | 27.2 | 25.0 | 21.2 | 26.6 | 24.3 | 20.7 | 34.0 | 33.5 | 52.3 | 50.8 | 28.4 | 3 |
| (32) | (1¼) | 158.8 | 43.2 | 43.2 | - | 43.7 | 35.5 | 32.9 | 29.9 | 35.1 | 32.5 | 29.5 | 42.7 | 42.2 | 63.5 | 63.5 | 28.4 | 5 |
| 40 | 1½ | 177.8 | 49.1 | 49.5 | - | 50.0 | 41.2 | 38.4 | 34.4 | 40.9 | 38.1 | 34.0 | 48.6 | 48.3 | 69.9 | 73.0 | 31.8 | 6 |
| 50 | 2 | 215.9 | 61.1 | 62.0 | - | 62.5 | 52.7 | 49.5 | 43.1 | 52.5 | 49.3 | 42.9 | 60.5 | 60.5 | 104.6 | 92.1 | 38.1 | 8 |
| 65 | 2½ | 244.3 | 77.1 | 74.7 | - | 75.4 | 65.9 | 62.3 | 57.3 | 62.7 | 59.0 | 54.0 | 76.3 | 73.2 | 124.0 | 104.8 | 41.1 | 8 |
| 80 | 3 | 241.3 | 90.0 | 90.7 | - | 91.4 | 78.1 | 73.9 | 66.9 | 77.9 | 73.7 | 66.9 | 89.1 | 88.9 | 127.9 | 127.9 | 38.1 | 10 |
| (90) | (3½) | - | - | - | - | - | 90.2 | 85.4 | 76.2 | 90.1 | 85.5 | - | - | - | - | - | - | - |
| 100 | 4 | 292.1 | 115.4 | 116.1 | - | 116.9 | 102.3 | 97.1 | 87.3 | 102.3 | 97.2 | 87.3 | 114.3 | 114.3 | 158.8 | 157.2 | 44.5 | 11 |
| (125) | (5) | 349.3 | 141.2 | 143.8 | - | 144.5 | 126.6 | 120.8 | 108.0 | 128.2 | 122.3 | 109.6 | 139.8 | 141.2 | 190.5 | 185.7 | 50.8 | 11 |
| 150 | 6 | 381 | 166.6 | 170.7 | - | 171.5 | 151.0 | 143.2 | 128.8 | 154.1 | 146.3 | 131.8 | 165.2 | 168.4 | 235.0 | 215.9 | 55.6 | 13 |
| 200 | 8 | 469.9 | 218.0 | 221.5 | - | 222.3 | 199.9 | 190.9 | 170.3 | 202.7 | 193.7 | 173.1 | 216.3 | 219.2 | 298.5 | 269.9 | 63.5 | 13 |
| 250 | 10 | 546.1 | 279.5 | 276.4 | - | 277.4 | 248.8 | 237.2 | 210.2 | 254.5 | 242.9 | 215.9 | 267.4 | 273.1 | 368.3 | 323.8 | 69.9 | 13 |
| 300 | 12 | 609.6 | 321.0 | 327.2 | - | 328.2 | 297.9 | 283.7 | 251.9 | 303.2 | 288.9 | 257.2 | 318.5 | 323.9 | 419.1 | 381.0 | 79.2 | 13 |
| 350 | 14 | 641.4 | 358.1 | 359.2 | - | 360.2 | 333.4 | 317.6 | 284.2 | 333.3 | 317.5 | 284.2 | 355.6 | 355.6 | 450.9 | 412.8 | 85.9 | 13 |
| 400 | 16 | 704.9 | 409.0 | 410.5 | - | 411.2 | 381.0 | 363.6 | 325.4 | 381.0 | 363.6 | 325.5 | 406.4 | 406.4 | 508.0 | 469.9 | 88.9 | 13 |
| 450 | 18 | 787.4 | 460.0 | 461.8 | - | 462.3 | 428.6 | 409.6 | 366.8 | 428.7 | 409.6 | 366.7 | 457.2 | 457.2 | 565.2 | 533.4 | 101.6 | 13 |
| 500 | 20 | 857.3 | 511.0 | 513.1 | - | 514.4 | 477.8 | 455.6 | 408.0 | 477.8 | 455.6 | 408.0 | 508.0 | 508.0 | 622.3 | 584.2 | 108.0 | 13 |
| 600 | 24 | 1041.4 | 613.0 | 616.0 | - | 616.0 | 574.6 | 547.8 | 490.6 | 574.6 | 547.6 | 490.6 | 609.6 | 609.6 | 749.3 | 692.2 | 139.7 | 13 |

# ANSI B16.5




## CLASS 600

Unit : mm

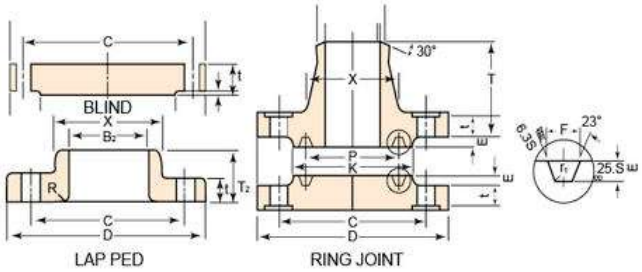
| Nominal Pipe Size | | Outside Dia. of Flanges | Dia. of Bore | | | | | | | | | | Dia. or Hub at Bevel A | | Dia. of Hub at Base | Dia. of Raised Face | thick of Flanges | Radius of Fillet |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Slip-on Socket $B_1$ | | Lap-Joint $B_2$ | | Welding-Neck, Socket $B_3$ | | | | | | | | | | | |
| | | | | | | | JPI | | | ANSI | | | | | | | | |
| A | B | D | JPI | ANSI | JPI | ANSI | SCH40 | SCH80 | SCH160 | SCH40 | SCH80 | SCH160 | JPI | ANSI | X | G | t | R |
| 15 | ½ | 95.3 | 22.2 | 22.4 | - | 22.9 | 16.1 | 14.3 | 12.3 | 15.8 | 13.9 | 11.8 | 21.7 | 21.3 | 38.1 | 34.9 | 14.3 | 3 |
| 20 | ¾ | 117.3 | 27.7 | 27.7 | - | 28.2 | 21.4 | 19.4 | 16.2 | 20.9 | 18.8 | 15.6 | 27.2 | 26.7 | 47.8 | 42.9 | 15.9 | 3 |
| 25 | 1 | 124.0 | 34.5 | 34.6 | - | 35.1 | 27.2 | 25.0 | 21.2 | 26.6 | 24.3 | 20.7 | 34.0 | 33.5 | 53.8 | 50.8 | 17.5 | 3 |
| (32) | (1¼) | 133.4 | 43.2 | 43.2 | - | 43.7 | 35.5 | 32.9 | 29.9 | 35.1 | 32.5 | 29.5 | 42.7 | 42.2 | 63.5 | 63.5 | 20.7 | 5 |
| 40 | 1½ | 155.4 | 49.1 | 49.6 | - | 50.0 | 41.2 | 38.4 | 34.4 | 40.9 | 38.1 | 34.0 | 48.6 | 48.3 | 69.9 | 73.0 | 22.3 | 6 |
| 50 | 2 | 165.1 | 61.1 | 62.0 | - | 62.5 | 52.7 | 49.5 | 43.1 | 52.5 | 49.3 | 42.9 | 60.5 | 60.5 | 84.1 | 92.1 | 25.4 | 8 |
| 65 | 2½ | 190.5 | 77.1 | 74.7 | - | 75.5 | 65.9 | 62.3 | 57.3 | 62.7 | 59.0 | 54.0 | 76.3 | 73.2 | 100.1 | 104.8 | 28.6 | 10 |
| 80 | 3 | 209.6 | 90.0 | 90.7 | - | 91.4 | 78.1 | 73.9 | 66.9 | 77.9 | 73.7 | 66.7 | 89.1 | 88.9 | 117.3 | 127.0 | 31.8 | 10 |
| (90) | (3½) | 228.6 | 102.6 | 103.4 | - | 104.2 | 90.2 | 85.4 | 76.2 | 90.1 | 85.5 | - | 101.6 | 101.6 | 133.4 | 139.7 | 35.0 | 11 |
| 100 | 4 | 273.1 | 115.4 | 116.1 | - | 116.9 | 102.3 | 97.1 | 87.3 | 102.3 | 97.2 | 87.3 | 114.3 | 114.3 | 152.4 | 157.2 | 38.1 | 11 |
| (125) | (5) | 330.2 | 141.2 | 143.8 | - | 144.6 | 126.6 | 120.8 | 108.0 | 128.2 | 122.3 | 109.6 | 139.8 | 141.2 | 189.0 | 185.7 | 44.5 | 13 |
| 150 | 6 | 355.6 | 166.6 | 170.7 | - | 171.5 | 151.0 | 143.2 | 128.8 | 154.1 | 146.3 | 131.8 | 165.2 | 168.4 | 222.3 | 215.9 | 47.7 | 13 |
| 200 | 8 | 419.1 | 218.0 | 221.5 | - | 222.4 | 199.9 | 190.9 | 170.3 | 202.7 | 193.7 | 173.1 | 216.3 | 219.2 | 273.1 | 269.9 | 55.6 | 13 |
| 250 | 10 | 508.0 | 269.5 | 276.4 | - | 277.4 | 248.8 | 237.2 | 210.2 | 254.5 | 242.9 | 215.9 | 267.4 | 273.1 | 342.9 | 323.8 | 63.5 | 13 |
| 300 | 12 | 558.8 | 321.0 | 327.2 | - | 328.2 | 297.9 | 283.7 | 251.9 | 303.2 | 288.9 | 257.2 | 318.5 | 323.9 | 400.1 | 381.0 | 66.7 | 13 |
| 350 | 14 | 603.3 | 358.1 | 359.2 | - | 360.2 | 333.4 | 317.6 | 284.2 | 333.3 | 317.5 | 284.2 | 355.6 | 355.6 | 431.8 | 412.8 | 69.9 | 13 |
| 400 | 16 | 685.8 | 409.0 | 410.2 | - | 411.3 | 381.0 | 363.6 | 325.4 | 281.0 | 363.6 | 325.5 | 406.4 | 406.4 | 495.3 | 469.9 | 76.2 | 13 |
| 450 | 18 | 743.0 | 460.0 | 461.8 | - | 462.3 | 428.6 | 409.6 | 368.8 | 428.7 | 409.6 | 266.7 | 457.2 | 457.2 | 546.1 | 533.4 | 82.6 | 13 |
| 500 | 20 | 812.8 | 511.0 | 513.1 | - | 514.4 | 477.8 | 455.6 | 408.0 | 477.8 | 455.6 | 408.0 | 508.0 | 508.0 | 609.6 | 584.2 | 88.9 | 13 |
| 600 | 24 | 939.8 | 613.0 | 616.0 | - | 616.0 | 574.6 | 547.8 | 490.6 | 574.6 | 547.7 | 490.6 | 609.6 | 609.6 | 717.6 | 692.2 | 101.6 | 13 |

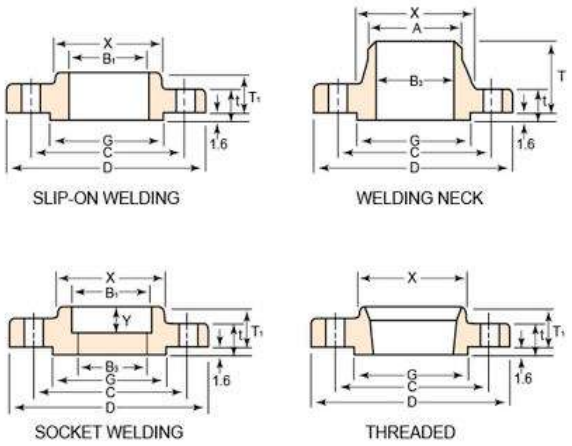
## CLASS 600

Unit : mm

| Nominal Pipe Size | | Length Thru Hub | | | Depth or Socket | Drilling | | | Dia. of Raised Face for Ring Joint | Pitch Dia. of Ring & Groove | Ring No | RTJ Type | | | Approx. Weight (kg) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Slip-on Socket Thread | Lap Joint | Welding Neck | | Dia. of Bolt Circle | Number or Bolt | Dia. of Bolt Hole | | | | Depth of Groove | Width of Groove | Radius of Groove | WN | SO | SW | BL |
| A | B | $T_1$ | $T_2$ | T | Y | C | N | | K (min) | P | | E | F | $r_1$ | | | | |
| 15 | ½ | 22.4 | 22.4 | 52.3 | 9.5 | 66.5 | 4 | 16 | 51.0 | 34.14 | R11 | 5.56 | 7.14 | 0.8 | 0.88 | 0.74 | 0.78 | 0.76 |
| 20 | ¾ | 25.4 | 25.4 | 57.2 | 11.1 | 82.6 | 4 | 19 | 63.5 | 42.88 | R13 | 6.35 | 8.74 | 0.8 | 1.44 | 1.24 | 1.29 | 1.27 |
| 25 | 1 | 26.9 | 27 | 62.0 | 12.7 | 88.9 | 4 | 19 | 70.0 | 50.80 | R16 | 6.35 | 8.74 | 0.8 | 1.81 | 1.51 | 1.58 | 1.59 |
| (32) | (1¼) | 28.4 | 29 | 66.5 | 14.3 | 98.6 | 4 | 19 | 79.5 | 60.32 | R18 | 6.35 | 8.74 | 0.8 | 2.49 | 2.03 | 2.14 | 2.24 |
| 40 | 1½ | 31.8 | 32 | 69.9 | 15.9 | 114.3 | 4 | 22 | 90.5 | 68.28 | R20 | 6.35 | 8.74 | 0.8 | 3.55 | 3.01 | 3.14 | 3.30 |
| 50 | 2 | 36.6 | 37 | 73.0 | 17.5 | 127.0 | 8 | 19 | 108 | 82.55 | R23 | 7.92 | 11.91 | 0.8 | 4.34 | 3.62 | 3.82 | 4.11 |
| 65 | 2½ | 41.1 | 41 | 79.2 | 19.0 | 149.4 | 8 | 22 | 127 | 101.60 | R26 | 7.92 | 11.91 | 0.8 | 6.41 | 5.20 | 5.57 | 6.20 |
| 80 | 3 | 46.0 | 46 | 82.6 | 20.6 | 168.1 | 8 | 22 | 146 | 123.82 | R31 | 7.92 | 11.91 | 0.8 | 8.62 | 7.08 | 7.59 | 8.51 |
| (90) | (3½) | 49.3 | 49.3 | 85.9 | 22.2 | 184.2 | 8 | 26 | 159 | 131.78 | R34 | 7.92 | 11.91 | 0.8 | 10.8 | 8.86 | - | 10.9 |
| 100 | 4 | 53.8 | 54 | 101.6 | 23.8 | 215.9 | 8 | 26 | 175 | 149.22 | R37 | 7.92 | 11.91 | 0.8 | 17.6 | 14.6 | - | 17.4 |
| (125) | (5) | 60.5 | 60.5 | 114.3 | 23.8 | 266.9 | 8 | 29 | 210 | 180.98 | R41 | 7.92 | 11.91 | 0.8 | 29.4 | 24.6 | - | 29.4 |
| 150 | 6 | 66.5 | 67 | 117.3 | 27.0 | 292.1 | 12 | 29 | 241 | 211.12 | R45 | 7.92 | 11.91 | 0.8 | 35.7 | 29.6 | - | 36.4 |
| 200 | 8 | 76.2 | 76 | 133.4 | 31.8 | 349.3 | 12 | 32 | 302 | 269.88 | R49 | 7.92 | 11.91 | 0.8 | 54.6 | 44.0 | - | 58.8 |
| 250 | 10 | 85.9 | 111.3 | 152.4 | 33.3 | 431.8 | 16 | 35 | 356 | 323.85 | R53 | 7.92 | 11.91 | 0.8 | 86.26 | 73.3 | - | 98.4 |
| 300 | 12 | 91.9 | 117.3 | 155.4 | 39.7 | 489.0 | 20 | 35 | 413 | 381.00 | R57 | 7.92 | 11.91 | 0.8 | 102.51 | 87.0 | - | 124 |
| 350 | 14 | 93.7 | 127.0 | 165.1 | 41.3 | 527.1 | 20 | 39 | 457 | 419.10 | R61 | 7.92 | 11.91 | 0.8 | 121.56 | 99.8 | - | 152 |
| 400 | 16 | 106.4 | 139.7 | 177.8 | 44.5 | 603.3 | 20 | 42 | 508 | 469.90 | R65 | 7.92 | 11.91 | 0.8 | 177.06 | 142 | - | 214 |
| 450 | 18 | 117.3 | 152.4 | 184.2 | 49.2 | 654.1 | 20 | 45 | 575 | 533.40 | R69 | 7.92 | 11.91 | 0.8 | 215.65 | 176 | - | 275 |
| 500 | 20 | 127.0 | 165.1 | 190.5 | 54.0 | 723.9 | 24 | 45 | 635 | 584.20 | R73 | 9.52 | 13.49 | 1.5 | 267.8 | 224 | - | 352 |
| 600 | 24 | 139.7 | 184.2 | 203.2 | 63.5 | 838.2 | 24 | 51 | 749 | 692.16 | R77 | 11.13 | 16.66 | 1.5 | 372.00 | 317 | - | 533.45 |

# ANSI B16.5




## CLASS 150

### ANSI & JPI FORGED FLANGES (JPI 7S-15-84 & ANSI B16.5)

Unit : mm

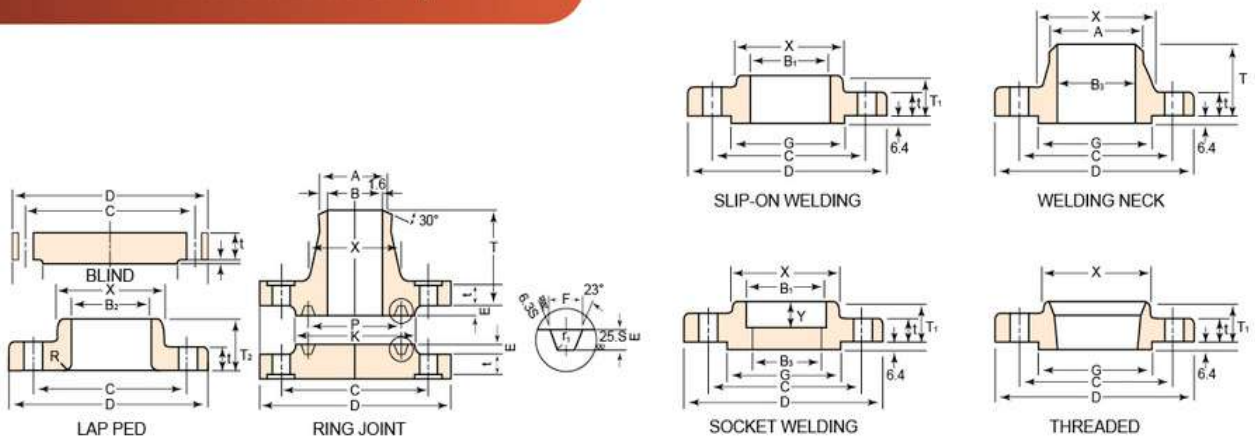
| Nominal Pipe Size | | Outside Dia. of Flanges | Dia. of Bore | | | | | | | | | | Dia. or Hub at Bevel A | | Dia. of Hub at Base | Dia. of Raised Face | thick of Flanges | Radius of Fillet |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Slip-on Socket $B_1$ | | Lap-Joint $B_2$ | | Welding-Neck, Socket $B_3$ | | | | | | | | | | | |
| | | | | | | | JPI | | | ANSI | | | | | | | | |
| A | B | D | JPI | ANSI | JPI | ANSI | SCH40 | SCH80 | SCH160 | SCH40 | SCH80 | SCH160 | JPI | ANSI | X | G | t | R |
| 15 | ½ | 88.9 | 22.2 | 22.4 | 23.4 | 22.9 | 16.1 | 14.3 | 12.3 | 15.8 | 13.9 | 11.8 | 21.7 | 21.3 | 30.2 | 35.1 | 11.2 | 3 |
| 20 | ¾ | 98.6 | 27.7 | 27.7 | 28.9 | 28.2 | 21.4 | 19.4 | 16.2 | 20.9 | 18.8 | 15.6 | 27.2 | 26.7 | 38.1 | 42.9 | 12.7 | 3 |
| 25 | 1 | 108.0 | 34.5 | 34.6 | 35.6 | 35.1 | 27.2 | 25.0 | 21.2 | 26.6 | 24.3 | 20.7 | 34.0 | 33.5 | 49.3 | 50.8 | 14.3 | 3 |
| (32) | (1¼) | 117.3 | 43.2 | 43.2 | 44.3 | 43.7 | 35.5 | 32.9 | 29.9 | 35.1 | 32.5 | 29.5 | 42.7 | 42.2 | 58.7 | 63.5 | 15.9 | 5 |
| 40 | 1½ | 127.0 | 49.1 | 49.6 | 50.4 | 50.0 | 41.2 | 38.4 | 34.4 | 40.9 | 38.1 | 34.0 | 48.6 | 48.3 | 65.0 | 73.2 | 17.5 | 6 |
| 50 | 2 | 152.4 | 61.1 | 62.0 | 62.7 | 62.5 | 52.7 | 49.5 | 43.1 | 52.5 | 49.3 | 42.9 | 60.5 | 60.5 | 77.8 | 91.9 | 19.1 | 8 |
| 65 | 2½ | 177.8 | 77.1 | 74.7 | 78.7 | 75.5 | 65.9 | 62.3 | 57.3 | 62.7 | 59.0 | 54.0 | 76.3 | 73.2 | 90.4 | 104.6 | 22.3 | 8 |
| 80 | 3 | 190.5 | 90.0 | 90.7 | 91.6 | 91.4 | 78.1 | 73.9 | 66.9 | 77.9 | 73.7 | 66.7 | 89.1 | 88.9 | 108.0 | 127.0 | 23.9 | 10 |
| (90) | (3½) | 215.9 | 102.6 | 103.4 | 104.1 | 104.2 | 90.2 | 85.4 | 76.2 | 90.1 | 85.5 | - | 101.6 | 101.6 | 122.2 | 139.7 | 23.9 | 10 |
| 100 | 4 | 228.6 | 115.4 | 116.1 | 116.9 | 116.9 | 102.3 | 97.1 | 87.3 | 102.3 | 97.2 | 87.3 | 114.3 | 114.3 | 134.9 | 157.2 | 23.9 | 11 |
| (125) | (5) | 254.0 | 141.2 | 143.8 | 143.0 | 144.6 | 126.6 | 130.8 | 108.0 | 128.2 | 122.3 | 109.6 | 139.8 | 141.2 | 163.6 | 185.7 | 23.9 | 11 |
| 150 | 6 | 279.4 | 166.6 | 170.7 | 168.4 | 171.5 | 151.0 | 143.2 | 128.8 | 154.1 | 146.3 | 131.8 | 165.2 | 168.4 | 192.0 | 215.9 | 25.4 | 13 |
| 200 | 8 | 342.9 | 218.0 | 221.5 | 219.5 | 222.3 | 199.9 | 190.9 | 170.3 | 202.7 | 193.7 | 173.1 | 216.3 | 219.2 | 246.1 | 269.7 | 28.6 | 13 |
| 250 | 10 | 406.4 | 269.5 | 276.4 | 271.7 | 277.4 | 248.8 | 237.2 | 210.2 | 254.5 | 242.9 | 215.9 | 267.4 | 273.1 | 304.8 | 323.9 | 30.2 | 13 |
| 300 | 12 | 482.6 | 321.0 | 327.2 | 322.8 | 328.2 | 297.9 | 283.7 | 251.9 | 303.2 | 288.9 | 257.2 | 318.5 | 323.9 | 365.3 | 381.0 | 31.8 | 13 |
| 350 | 14 | 533.4 | 358.1 | 359.2 | - | 360.2 | 333.4 | 317.6 | 284.2 | 333.6 | 317.5 | 284.2 | 355.6 | 355.6 | 400.1 | 412.8 | 35.1 | 13 |
| 400 | 16 | 596.9 | 409.0 | 410.5 | - | 411.3 | 381.0 | 363.6 | 325.4 | 381.0 | 363.6 | 325.5 | 406.4 | 406.4 | 457.2 | 469.9 | 36.6 | 13 |
| 450 | 18 | 635.0 | 460.0 | 461.8 | - | 462.4 | 428.6 | 409.6 | 366.8 | 428.7 | 409.6 | 366.7 | 457.2 | 457.2 | 504.8 | 533.4 | 39.6 | 13 |
| 500 | 20 | 698.5 | 511.0 | 513.1 | - | 514.4 | 477.8 | 455.6 | 408.0 | 477.8 | 455.6 | 408.0 | 508.0 | 508.0 | 558.8 | 584.2 | 42.9 | 13 |
| 600 | 24 | 812.8 | 613.0 | 616.0 | - | 616.0 | 574.6 | 547.8 | 490.6 | 574.7 | 547.7 | 490.6 | 609.6 | 609.6 | 663.4 | 692.2 | 47.7 | 13 |

## CLASS 150

Unit : mm

| Nominal Pipe Size | | Length Thru Hub | | | Depth or Socket | Drilling | | | Dia. of Raised Face for Ring Joint | Pitch Dia. of Ring & Groove | Ring No | RTJ Type | | | Approx. Weight (kg) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Slip-on Socket Thread | Lap Joint | Welding Neck | | Dia. of Bolt Circle | Number or Bolt | Dia. of Bolt Hole | | | | Depth of Groove | Width of Groove | Radius of Groove | WN | SO | SW | BL |
| A | B | $T_1$ | $T_2$ | T | Y | C | N | | K (min) | P | | E | F | $r_1$ | | | | |
| 15 | ½ | 15.9 | 15.9 | 47.6 | 9.5 | 60.5 | 4 | 16 | - | - | - | - | - | - | 0.50 | 0.41 | 0.42 | 0.43 |
| 20 | ¾ | 15.9 | 15.9 | 52.4 | 11.1 | 70.0 | 4 | 16 | - | - | - | - | - | - | 0.75 | 0.57 | 0.58 | 0.62 |
| 25 | 1 | 17.5 | 17.5 | 55.6 | 12.7 | 79.2 | 4 | 16 | 63.5 | 47.62 | R15 | 6.35 | 8.74 | 0.8 | 1.02 | 0.78 | 0.80 | 0.87 |
| (32) | (1¼) | 20.6 | 20.6 | 57.1 | 14.3 | 89.0 | 4 | 16 | 73.0 | 57.15 | R17 | 6.35 | 8.74 | 0.8 | 1.33 | 1.03 | 1.06 | 1.16 |
| 40 | 1½ | 22.2 | 22.2 | 61.9 | 15.9 | 98.5 | 4 | 16 | 82.5 | 69.07 | R19 | 6.35 | 8.74 | 0.8 | 1.72 | 1.32 | 1.35 | 1.54 |
| 50 | 2 | 25.4 | 25.4 | 63.5 | 17.5 | 120.5 | 4 | 19 | 102 | 82.55 | R22 | 6.35 | 8.74 | 0.8 | 2.59 | 2.09 | 2.13 | 2.46 |
| 65 | 2½ | 28.6 | 28.6 | 69.8 | 19.0 | 139.5 | 4 | 19 | 121 | 101.60 | R25 | 6.35 | 8.74 | 0.8 | 4.06 | 3.25 | 3.35 | 3.99 |
| 80 | 3 | 30.2 | 30.2 | 69.8 | 20.6 | 152.5 | 4 | 19 | 133 | 114.30 | R29 | 6.35 | 8.74 | 0.8 | 4.95 | 3.43 | 4.02 | 4.98 |
| (90) | (3½) | 31.8 | 31.8 | 71.4 | 22.2 | 178.0 | 8 | 19 | 154 | 131.78 | R33 | 6.35 | 8.74 | 0.8 | 6.08 | 4.86 | 4.99 | 6.20 |
| 100 | 4 | 33.3 | 33.3 | 76.2 | 23.8 | 190.5 | 8 | 19 | 171 | 149.22 | R36 | 6.35 | 8.74 | 0.8 | 6.91 | 5.35 | 5.99 | 7.05 |
| (125) | (5) | 36.5 | 36.5 | 88.9 | 23.8 | 216.0 | 8 | 22 | 194 | 171.45 | R40 | 6.35 | 8.74 | 0.8 | 8.75 | 6.22 | 6.68 | 8.67 |
| 150 | 6 | 39.7 | 39.7 | 88.9 | 27.0 | 241.5 | 8 | 22 | 219 | 193.68 | R43 | 6.35 | 8.74 | 0.8 | 10.8 | 7.38 | 7.99 | 11.3 |
| 200 | 8 | 44.5 | 44.5 | 101.6 | 31.8 | 298.5 | 8 | 22 | 273 | 247.65 | R48 | 6.35 | 8.74 | 0.8 | 17.7 | 12.36 | 13.29 | 19.9 |
| 250 | 10 | 49.2 | 49.2 | 101.6 | 33.3 | 362.0 | 12 | 25 | 330 | 304.80 | R52 | 6.35 | 8.74 | 0.8 | 24.9 | 17.10 | 19.50 | 29.0 |
| 300 | 12 | 55.6 | 55.6 | 114.3 | 39.7 | 432.0 | 12 | 25 | 406 | 381.00 | R56 | 6.35 | 8.74 | 0.8 | 38.6 | 27.68 | 29.03 | 43.7 |
| 350 | 14 | 57.2 | 79.3 | 127.0 | 41.3 | 476.0 | 12 | 29 | 425 | 396.88 | R59 | 6.35 | 8.74 | 0.8 | 50.6 | 35.2 | 38.56 | 58.6 |
| 400 | 16 | 63.5 | 87.4 | 127.0 | 44.5 | 539.5 | 16 | 29 | 483 | 454.02 | R64 | 6.35 | 8.74 | 0.8 | 63.9 | 42.18 | 44.49 | 76.6 |
| 450 | 18 | 68.3 | 96.8 | 139.7 | 49.2 | 578.0 | 16 | 32 | 546 | 517.52 | R68 | 6.35 | 8.74 | 0.8 | 74.9 | 49.7 | 54.43 | 94.5 |
| 500 | 20 | 73.0 | 103.1 | 144.5 | 54.0 | 635.0 | 20 | 32 | 597 | 558.80 | R72 | 6.35 | 8.74 | 0.8 | 93.3 | 63.5 | 70.31 | 123 |
| 600 | 24 | 82.6 | 111.3 | 152.4 | 63.5 | 749.5 | 20 | 35 | 711 | 673.10 | R76 | 6.35 | 8.74 | 0.8 | 119.7 | 90.5 | 95.25 | 188 |

# ANSI B16.5




## CLASS 2500

Unit : mm

| Nominal Pipe Size | | Outside Dia. of Flanges | Dia. of Bore | | | | | | | | | | Dia. or Hub at Bevel A | | Dia. of Hub at Base | Dia. of Raised Face | thick of Flanges | Radius of Fillet |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Slip-on Socket $B_3$ | | Lap-Joint $B_2$ | | Welding-Neck, Socket $B_3$ | | | | | | | | | | | |
| | | | | | | | JPI | | | ANSI | | | | | | | | |
| A | B | D | JPI | ANSI | JPI | ANSI | SCH40 | SCH80 | SCH160 | SCH40 | SCH80 | SCH160 | JPI | ANSI | X | G | t | R |
| 15 | ½ | 133.4 | 22.2 | 22.4 | - | 22.9 | 16.1 | 14.3 | 12.3 | 15.8 | 13.9 | 11.8 | 21.7 | 21.3 | 42.9 | 34.9 | 30.2 | 3 |
| 20 | ¾ | 139.7 | 27.7 | 27.7 | - | 28.2 | 21.4 | 19.4 | 16.2 | 20.9 | 18.8 | 15.6 | 27.2 | 26.7 | 50.8 | 42.9 | 31.8 | 3 |
| 25 | 1 | 158.8 | 34.5 | 34.5 | - | 35.1 | 27.2 | 25.0 | 21.2 | 26.6 | 24.3 | 20.7 | 34.0 | 33.5 | 57.2 | 50.8 | 35.1 | 3 |
| (32) | (1¼) | 184.2 | 43.2 | 43.2 | - | 43.7 | 35.5 | 32.9 | 29.2 | 35.1 | 32.5 | 29.5 | 42.7 | 42.2 | 73.2 | 63.5 | 38.1 | 5 |
| 40 | 1½ | 203.2 | 49.1 | 49.5 | - | 50.0 | 41.2 | 38.4 | 34.4 | 40.9 | 38.1 | 34.0 | 48.6 | 48.3 | 79.2 | 73.0 | 44.5 | 6 |
| 50 | 2 | 235 | 61.1 | 62.0 | - | 62.5 | 52.7 | 49.5 | 43.1 | 52.5 | 49.3 | 42.9 | 60.5 | 60.5 | 95.3 | 92.1 | 50.8 | 8 |
| 65 | 2½ | 266.7 | 77.1 | 74.7 | - | 75.4 | 65.9 | 62.3 | 57.3 | 62.7 | 59.0 | 54.0 | 76.3 | 73.2 | 114.3 | 104.8 | 57.2 | 8 |
| 80 | 3 | 304.8 | 90.0 | 90.7 | - | 91.4 | 78.1 | 73.9 | 66.9 | 77.9 | 73.7 | 66.9 | 89.1 | 88.9 | 133.4 | 127.0 | 66.5 | 10 |
| (90) | (3½) | - | - | - | - | - | 90.2 | 85.4 | 76.2 | 90.1 | 85.5 | - | - | - | - | - | - | - |
| 100 | 4 | 355.6 | 115.4 | 116.1 | - | 116.9 | 102.3 | 97.1 | 87.3 | 102.3 | 97.2 | 87.3 | 114.3 | 114.3 | 165.1 | 157.2 | 76.2 | 11 |
| (125) | (5) | 419.1 | 141.2 | 143.8 | - | 144.5 | 126.6 | 120.8 | 108.0 | 128.2 | 122.3 | 109.6 | 139.8 | 141.2 | 203.2 | 185.7 | 91.9 | 11 |
| 150 | 6 | 482.6 | 166.6 | 170.7 | - | 171.5 | 151.0 | 143.2 | 128.8 | 154.1 | 146.3 | 131.8 | 165.2 | 168.4 | 235.0 | 215.9 | 108.0 | 13 |
| 200 | 8 | 552.5 | 218.0 | 221.5 | - | 222.3 | 199.9 | 190.9 | 170.3 | 202.7 | 193.7 | 173.1 | 216.3 | 219.2 | 304.8 | 269.9 | 127.0 | 13 |
| 250 | 10 | 673.1 | 269.5 | 276.4 | - | 277.4 | 248.8 | 237.2 | 210.2 | 254.5 | 242.9 | 215.9 | 267.4 | 273.1 | 374.7 | 323.8 | 165.1 | 13 |
| 300 | 12 | 762 | 321.0 | 327.2 | - | 328.2 | 297.9 | 283.7 | 251.9 | 303.2 | 288.9 | 257.2 | 318.5 | 323.9 | 441.5 | 381.0 | 184.2 | 13 |

## CLASS 2500

Unit : mm

| Nominal Pipe Size | | Length Thru Hub | | | Depth or Socket | Drilling | | | Dia. of Raised Face for Ring Joint | Pitch Dia. of Ring & Groove | Ring No | RTJ Type | | | Approx. Weight (kg) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Slip-on Socket Thread | Lap Joint | Welding Neck | | Dia. of Bolt Circle | Number or Bolt | Dia. of Bolt Hole | | | | Depth of Groove | Width of Groove | Radius of Groove | WN | SO | SW | BL |
| A | B | $T_1$ | $T_2$ | T | Y | C | N | | K (min) | P | | E | F | $r_1$ | | | | |
| 15 | ½ | 39.6 | 39.6 | 73.2 | 9.5 | 88.9 | 4 | 23 | 60.5 | 42.88 | R13 | 6.35 | 8.74 | 0.8 | 3.23 | 3.2 | - | 3.09 |
| 20 | ¾ | 42.9 | 42.9 | 79.2 | 11.1 | 95.3 | 4 | 23 | 73.0 | 50.80 | R16 | 6.35 | 8.74 | 0.8 | 3.86 | 3.6 | - | 3.66 |
| 25 | 1 | 47.8 | 47.8 | 88.9 | 12.7 | 108.0 | 4 | 26 | 82.5 | 60.32 | R18 | 6.35 | 8.74 | 0.8 | 5.40 | 5.0 | - | 5.14 |
| (32) | (1¼) | 52.3 | 52.3 | 95.3 | 14.3 | 130.0 | 4 | 29 | 102 | 72.24 | R21 | 7.92 | 11.91 | 0.8 | 8.13 | 7.3 | - | 7.74 |
| 40 | 1½ | 60.5 | 60.5 | 111.3 | 15.9 | 146.1 | 4 | 32 | 114 | 82.55 | R23 | 7.92 | 11.91 | 0.8 | 11.3 | 10.0 | - | 10.8 |
| 50 | 2 | 69.9 | 69.9 | 127.0 | 19.5 | 171.5 | 8 | 29 | 133 | 101.60 | R26 | 7.92 | 11.91 | 0.8 | 16.7 | 17.2 | - | 16.1 |
| 65 | 2½ | 79.2 | 79.2 | 142.7 | 19.0 | 196.9 | 8 | 32 | 149 | 111.12 | R28 | 9.52 | 13.49 | 1.5 | 24.5 | 24.9 | - | 23.7 |
| 80 | 3 | 91.9 | 91.9 | 168.1 | 20.6 | 228.6 | 8 | 35 | 168 | 127.00 | R32 | 9.52 | 13.49 | 1.5 | 37.4 | 37.6 | - | 36.0 |
| (90) | (3½) | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 100 | 4 | 108 | 108 | 190.5 | 23.8 | 273.1 | 8 | 42 | 203 | 157.18 | R38 | 11.13 | 16.66 | 1.5 | 57.3 | 57.7 | - | 55.9 |
| (125) | (5) | 130 | 130 | 228.6 | 23.8 | 323.9 | 8 | 48 | 241 | 190.50 | R42 | 12.70 | 19.84 | 1.5 | 96.1 | 95.3 | - | 94.1 |
| 150 | 6 | 152.4 | 152.4 | 273.1 | 27.0 | 368.3 | 8 | 54 | 279 | 228.60 | R47 | 12.70 | 19.84 | 1.5 | 149 | 146.57 | - | 146 |
| 200 | 8 | 177.8 | 177.8 | 317.5 | 31.8 | 438.2 | 12 | 54 | 340 | 279.40 | R51 | 14.27 | 23.01 | 1.5 | 222 | 220.0 | - | 220 |
| 250 | 10 | 228.6 | 228.6 | 419.1 | 33.3 | 640.0 | 12 | 67 | 425 | 342.90 | R55 | 17.48 | 30.18 | 2.4 | 434 | 419.57 | - | 429 |
| 300 | 12 | 254.0 | 254.0 | 463.6 | 39.7 | 619.3 | 12 | 74 | 495 | 406.40 | R60 | 17.48 | 33.32 | 2.4 | 610 | 499.0 | - | 609 |